

Adaptive Kernel Selection for Stein Variational Gradient Descent

Anonymous Authors

Abstract

This paper foregrounds kernel selection as a primary design choice in Stein Variational Gradient Descent (SVGD). SVGD transports particles toward a Bayesian posterior through update directions in a reproducing kernel Hilbert space (RKHS), and its convergence and approximation quality therefore depend strongly on the selected kernel. The standard median heuristic gives a convenient bandwidth rule, but it is inflexible and can perform poorly, especially in higher-dimensional problems. Our first contribution is an adaptive kernel-selection procedure over a parameterized kernel family: Adaptive SVGD (Ad-SVGD) alternates ordinary SVGD particle transport with gradient-ascent updates of kernel parameters that maximize the kernelized Stein discrepancy (KSD). Our second contribution is a mean-field convergence analysis that extends fixed-kernel KSD arguments to this adaptive setting and establishes convergence of the maximal KSD over the kernel class, with refined KL bounds under a Stein log-Sobolev condition. Our third contribution is an empirical evaluation showing that Ad-SVGD improves over median-heuristic SVGD across toy, inverse-problem, and Bayesian logistic-regression settings, including cases where adaptive bandwidths reduce variance underestimation.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle method for approximate Bayesian inference and a common alternative to traditional Markov Chain Monte Carlo (MCMC) methods. Its appeal comes from transporting an ensemble of particles along directions derived from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with those directions constrained by the unit ball of a reproducing kernel Hilbert space (RKHS). This paper focuses on the key consequence of that construction: the kernel is not a secondary implementation detail, but a central determinant of the particle interaction, convergence behavior, and posterior approximation quality.

The contribution we develop is an adaptive kernel-selection mechanism for the original SVGD dynamics. Existing evidence shows that SVGD is highly sensitive to the kernel choice [8, 28] and that its particle approximation often underestimates posterior variance [2]. These shortcomings have supported the view that SVGD becomes unreliable as the dimension of the state space grows. We revisit that view by asking how far the base SVGD method can be pushed when the kernel parameters are selected during inference by maximizing the kernelized Stein discrepancy (KSD).

1.1 Related work.

Since its introduction [18], SVGD has become a widely used inference tool across machine learning applications [22, 24, 30, 15]. The theoretical literature now provides several lenses for understanding the method: mean-field convergence has been studied in continuous time [23, 8, 5] and discrete

time [16, 32], finite-particle convergence rates have been established under different assumptions [3, 33], and the dynamics have been interpreted as gradient flows on spaces of probability distributions [19, 8], in analogy with Jordan, Kinderlehrer, and Otto [14]. Our work uses this theory not to propose a new transport architecture, but to make the kernel choice itself the object of optimization.

Kernel choice is therefore the specific gap addressed in this paper. The kernel determines particle interactions and affects the convergence behavior of the method (see also Figures 1 and 9). Moreover, standard convergence statements are often phrased in terms of KSD, and whether KSD convergence implies weak convergence depends on the selected kernel [11]. The median heuristic [12] is easy to implement, but it has limited theoretical support and is known to degrade as dimensionality increases [2, 40, 36]. Other lines of work address high-dimensional degradation through modified SVGD variants [7, 37, 10, 21] or neural-network parameterizations of the update direction [17, 39]; our contribution is complementary because it isolates the effect of adaptively choosing kernels within the original SVGD update.

The closest prior contribution is the mixture-of-kernels framework of Ai et al. [1]. That method defines a KSD for a weighted linear combination of kernels and learns the kernel weights by maximizing this multiple-kernel KSD. We retain the same high-level principle of selecting kernels through KSD maximization, but our contribution shifts the setting from finite kernel bases to continuous kernel-parameter optimization.

1.2 Contributions.

We organize the paper around one reviewer-facing question: can the original SVGD method, without architectural changes to the particle update, become substantially more effective when the kernel is selected adaptively and principled by KSD maximization? By separating kernel learning from other SVGD modifications, we make the role of kernel choice explicit and evaluate how much of the observed performance gap can be closed by adaptive bandwidth selection alone.

The paper makes three concrete contributions.

- (i) **Adaptive kernel-selection method.** We introduce Ad-SVG, which updates SVGD particles while adapting continuous kernel parameters by maximizing the KSD. Unlike the median heuristic, which supplies a fixed scalar bandwidth rule, the method supports multiple learned bandwidths and thereby makes kernel adaptivity part of the inference loop.
- (ii) **Mean-field convergence analysis.** We extend fixed-kernel discrete-time SVGD arguments to the adaptive setting. The main theoretical statement shows that the supremum of the KSD over the parameterized kernel class converges to zero, and a refined result gives mean-field iteration-complexity bounds under a Stein logarithmic Sobolev inequality.
- (iii) **Empirical validation of the kernel contribution.** Across numerical examples, we compare the adaptive rule directly against median-heuristic SVGD and show improved posterior approximation, including reduced variance collapse in higher-dimensional and Bayesian inference tasks.

2 Mathematical background

We first fix the notation needed to state the kernel-selection contribution precisely. Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a symmetric positive definite kernel with associated RKHS $\mathcal{H}_0$. We define $\mathcal{H}$ as the

d -fold Cartesian product of $\mathcal{H}_0$ equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ is the subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we use the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , namely probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify discrepancies between measures through the largest deviation of test-function integrals. For our purposes, their importance is that Stein's method [35] turns this idea into a computable diagnostic against an intractable target: it constructs test functions with zero mean under the target distribution. For the operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. This leads to the *kernelized Stein discrepancy* (KSD), the quantity that our method maximizes over kernel parameters:

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

Here $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The optimizer is $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map associated with k . Thus the KSD is the RKHS norm $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$, making explicit why changing the kernel changes both the discrepancy being minimized and the SVGD direction.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and reference distribution μ_0 , SVGD constructs an approximation by repeatedly pushing μ_n through $T(x) = x + \gamma \psi^{\mu_n}(x)$ and setting $\mu_{n+1} := T_{\#} \mu_n$. The vector field $\psi^\mu = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the steepest descent direction induced by the chosen RKHS. This link is the mechanism our contribution exploits: the solution of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_{\#} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes the KL-divergence. For this direction,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\#} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these kernel-dependent maps to particles $\{X_0^i\}_{i=1}^M$ sampled from μ_0 yields the standard SVGD update used as the base transport step throughout the paper:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

The method contribution starts from a limitation in existing theory. Because proving SVGD convergence in KL divergence is difficult and often requires restrictive assumptions, recent analyses instead study convergence through KSD [16, 33, 32]. KSD convergence by itself, however, does not automatically imply weak convergence: the mean-field measures may fail to be tight, which is necessary for weak convergence on Polish spaces. In addition, Equation (1) shows that the RKHS—and hence the kernel—controls how KSD convergence relates to weak convergence [11].

Ad-SVGd makes this dependency explicit by replacing a single fixed kernel with a parameterized family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, let KSD_θ be the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the optimal update direction, and Φ_θ the associated feature map. The purpose of this parameterization is reviewer-facing and practical: it turns kernel choice into an optimization variable that can be updated during inference instead of chosen once by a heuristic.

The selection criterion is to maximize the KSD between the current particle distribution and the target. This criterion is not merely diagnostic: the instantaneous KL decrease under SVGD is proportional to the squared KSD at the current particle measure (see Equations (2) and (7)). Therefore, selecting the kernel with the largest KSD targets the largest local KL decrease available within the kernel family. Geometric analyses of SVGD as a gradient flow support the same framing, showing that kernels with larger KSD values can yield more favorable convergence properties when their RKHSs are compared [28, 8].

The algorithmic contribution is thus to target the largest KSD over the kernel class, rather than the KSD associated with one fixed kernel. In contrast to the median heuristic of Liu and Wang [18], this formulation applies to any parameterized kernel family: at iteration n , the ideal parameter satisfies

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

In practice, Ad-SVGd approximates this maximization by taking one or more gradient-ascent steps with step size $s > 0$ on the KSD before the particle update (see Algorithm 1). To reduce runtime, the user may skip parameter updates on selected SVGD iterations through the decision variable `paramupdate`. This modular design is part of the contribution: the particle transport step can be replaced by adaptive step sizes, line searches, momentum, or second-order/preconditioned updates without changing the kernel-selection mechanism, and the gradient ascent on the kernel parameter can likewise be replaced by another optimizer. Our implementation uses the representation [6, 20]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k \, d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the parameter gradient is computed directly. The additional cost relative to vanilla SVGD is controlled by two implementation choices:

- The kernel-parameter ascent reuses the same gradients $\nabla \log \pi(X_n^i)$ as the corresponding SVGD step, so Ad-SVGd does not require additional evaluations of $\log \pi$ gradients.

- The cost can be reduced by using a small number of parameter steps $n_{\text{stepsttheta}}$, by updating the kernel only after several SVGD steps, or, for large particle sets, by estimating the KSD with subsampled particles.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $n_{\text{steps}}, n_{\text{stepsttheta}} \in \mathbb{N}$
Output: Final particle set $\{X_{n_{\text{steps}}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to  $n_{\text{steps}} - 1$  do
    Decide  $\text{paramupdate} \in \{\text{True}, \text{False}\}$ ;
    if  $\text{paramupdate}$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $n_{\text{stepsttheta}} - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{n_{\text{stepsttheta}}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

The theoretical contribution is a mean-field convergence analysis for the adaptive kernel choice. We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$ and extend recent KSD convergence results for fixed-kernel SVGD. Specifically, we adapt the analysis of [32] under Talagrand's inequality, which holds under mild assumptions on the target distribution and is weaker than the commonly used logarithmic Sobolev inequality. The assumptions below are the same type of assumptions as in [32], imposed uniformly over all kernel parameters so that the adaptive maximization remains covered.

Assumption 1. We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality $T1$, which means that there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

We first restate the fixed-kernel descent tool that the adaptive result builds on. Throughout this section, let

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

The following fundamental inequality supplies the fixed-kernel building block. For a fixed kernel parameter $\theta \in \Theta$ satisfying Assumption 3, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [32], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The main technical step behind (7) is verifying $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ through Talagrand's inequality, Assumption 2, which permits the use of (6) with $g = \psi_\theta^{\mu_n}$. Condition (7) then implies the fixed-kernel benchmark

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$.

With adaptive parameters (θ_n) , the inequality (7) becomes the same descent mechanism with the currently selected kernel:

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

We now state the adaptive analogue, first in the idealized case where the KSD maximization over the kernel family is solved exactly. Suppose the adaptive SVGD iteration has the simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

so that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

This exact formulation is available when the KSD maximization over the kernel parameter has a closed-form solution. One example is the multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For continuous kernel families, the update θ_n in (9) is generally approximated rather than solved in closed form. In the exact statement below, we assume $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$. The resulting descent condition is a direct consequence of Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The practically relevant case is approximate KSD maximization. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

where $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ denotes an iterative rule that aims to maximize $\operatorname{KSD}_\theta(\mu_n | \pi)$. In Algorithm 1, Ψ_n is gradient ascent on the KSD, but the analysis only needs the approximation behavior stated next and can therefore accommodate other suitable optimizers.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) - \operatorname{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

This assumption yields the main adaptive convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

The refined theoretical contribution appears in Section B. Under a Stein logarithmic Sobolev inequality, we strengthen the adaptive KSD result to KL convergence and derive iteration-complexity bounds for the mean-field system (10) under specific regimes for ε_n . The same analysis also identifies the bias induced when the KSD maximization is solved only to fixed $\bar{\varepsilon}$ accuracy at each iteration. Importantly, the adaptive formulation only requires one kernel in the class $\{k_\theta\}$ to satisfy the Stein logarithmic Sobolev inequality, which gives additional flexibility for verifying the condition.

5 Numerical Experiments

The experiments are designed to isolate and test the paper’s central contribution: adaptive kernel selection within otherwise standard SVGD. We first use a one-dimensional Gaussian mixture to show the bandwidth sensitivity that motivates the method, then evaluate Ad-SVGd on a linear inverse problem governed by an ordinary differential equation. Additional experiments in Section A examine dimensional scaling, and Section 5.4 compares Ad-SVGd with median-heuristic SVGd on the Bayesian logistic regression benchmark used in prior SVGd work.

5.1 Kernel parameterization

We use the experimental setup to make the kernel-selection question concrete. SVGd is most commonly implemented with kernels

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. The parameter h is the kernel *bandwidth*, and the median heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the current median particle distance. This rule is intended to balance the two terms in the SVGd update (3) [18], but it supplies only a heuristic bandwidth choice rather than an optimized one.

Ad-SVGd uses this flexibility through product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with $h = (h_1, \dots, h_d)$, allowing a distinct bandwidth in each dimension. The derivatives required by Algorithm 1 for these kernels (see Equations (4) and (5)) are straightforward to compute. We also tested a component-wise median heuristic for the same product kernels, using a naive median per dimension, but this did not perform well and suffered from variance collapse.

This parameterization is not the only possible use of the contribution. Beyond the experiments reported here, Ad-SVGD can be paired with more scalable kernel families in higher dimensions: spectral or low-rank parameterizations could adapt dominant directions of variability without the exponential cost of full product kernels, while Matérn or mixed-product kernels could add control over smoothness and anisotropy.

5.2 Toy example

We begin with the one-dimensional example from Liu and Wang [18] because it cleanly exposes the bandwidth-selection problem. The target is the Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles initialized from $\mathcal{N}(0, 1)$ and run SVGD for 10^4 steps with step size 1, using the kernels above with $p = 1$ and several fixed bandwidths h .

Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) against an exact sample of size 10^5 . Figure 1 shows the main motivation for Ad-SVGD: fixed-bandwidth SVGD performs well only over a limited range of h , so bandwidth selection materially affects the final approximation.

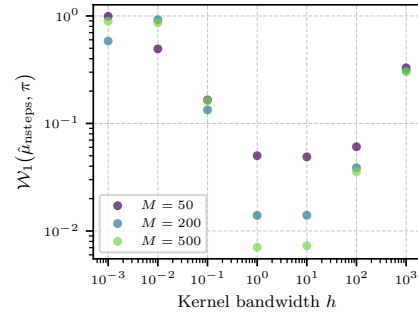


Figure 1: Bandwidth sensitivity motivating adaptive selection: final Wasserstein 1-distances for the one-dimensional example using SVGD with different fixed bandwidths h .

5.3 Linear inverse problem based on ODE

The second experiment tests whether adaptive kernel selection improves posterior uncertainty in a structured inverse problem. The setup is adapted from [38, Example 2.1]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

The associated inverse problem is to recover the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, we place a Gaussian process (GP) prior on u through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$.

The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For implementation, we replace H by a numerical discretization of Equation (11) on a grid with mesh size 2^{-8} and use the fully observed system, so $N_{\text{obs}} = 2^8$. We retain $N_x = 16$ KL terms for u , set the noise covariance to $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$, and construct reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

The comparison isolates the contribution of adaptive bandwidths: Med-SVGD uses the median heuristic, while Ad-SVGD uses the kernels from Section 5.1 with $p = 1$. We test particle ensemble sizes $M \in \{50, 100, 200\}$ and run both algorithms for $4 \cdot 10^5$ iterations with particle step size 10^{-3} and, as suggested by Liu and Wang [18], an AdaGrad variant for adaptive step-size control. For Ad-SVGD, the bandwidth step size is 10^{-5} and the bandwidth is updated only once every 100 particle updates. Under this scheme, Med-SVGD and Ad-SVGD had no significant runtime difference.

Figure 2 summarizes the qualitative result for an exemplary seed: both methods approximate the posterior mean, but Ad-SVGD more accurately captures posterior uncertainty. To quantify this contribution, we compute the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the particle sample mean and covariance [29, Section 3.2]. Since π is multivariate normal, this distance has the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the posterior mean and covariance. We also compare marginal variances of the final particle distribution with the posterior variances. Figure 3 aggregates 56 random seeds and shows the main empirical contribution in this inverse problem: Ad-SVGD gives a better posterior approximation than Med-SVGD, which underestimates uncertainty. Unlike Med-SVGD, Ad-SVGD also improves when the particle count increases beyond 50. Figure 4 explains how the adaptive mechanism behaves: component-wise bandwidths stabilize faster than the approximation error, differ clearly across components, and end up negatively correlated with the corresponding marginal variances.

5.4 Bayesian logistic regression

The Bayesian logistic regression (BLR) experiment tests the adaptive-kernel contribution on a benchmark used in earlier SVGD work. Following Liu and Wang [18] and Liu et al. [21], we use the

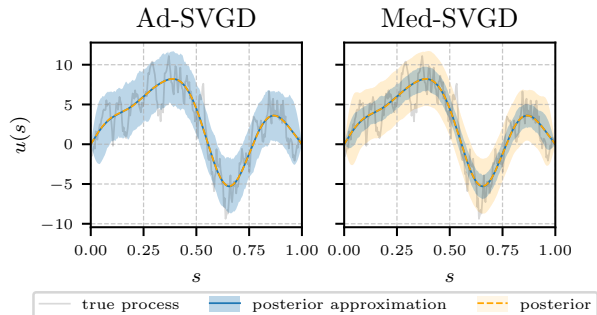
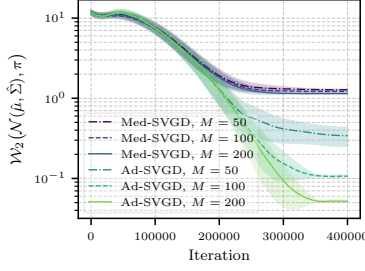
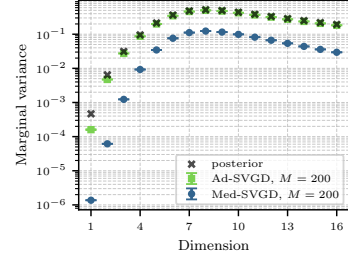


Figure 2: Posterior reconstruction contribution in the ODE-based inverse problem: GP mean and 90% confidence interval.

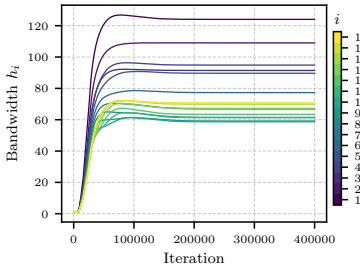


(a) Approximation error over iterations

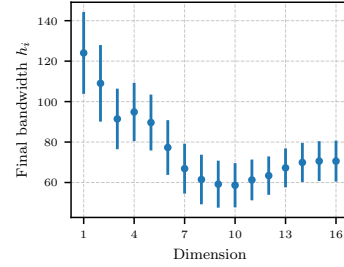


(b) Posterior marginal variance recovery

Figure 3: Empirical contribution on the ODE-based inverse problem: aggregated mean and 95% confidence interval over 56 random seeds comparing Med-SVGD and Ad-SVGD.



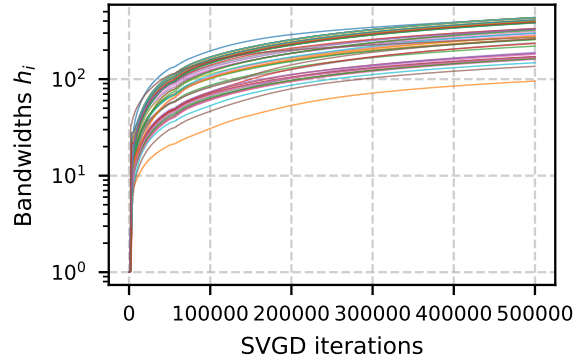
(a) Mean bandwidth evolution per dimension



(b) Final bandwidths with 95% confidence intervals

Figure 4: Adaptive mechanism in the ODE-based inverse problem: bandwidth behavior for Ad-SVGD aggregated over 56 random seeds.

Covertypes data set [4]. For regression weights w , we assign the Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$ and infer the posterior of $x = [w, \alpha]$. As in Liu et al. [21], we generate a reference with the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] and use it to evaluate the posterior approximation quality of Med-SVGD and Ad-SVGD. Following Liu et al. [21], we evaluate on subsets of 1000 observations from the original data set and aggregate over 10 random draws. Classification performance is measured by test-set prediction accuracy using the particle mean. Figure 5 shows that Ad-SVGD outperforms Med-SVGD for almost all seeds, achieving higher test accuracy while remaining closer to the MCMC reference. The same figure also reports squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples as a posterior-approximation metric. The methods differ substantially in how well they capture the posterior. Figure 7, using the visualization style of Liu et al. [21], compares final-particle covariance matrices with the MCMC reference averaged over the 10 seeds: Med-SVGD severely underestimates posterior uncer-



11
Figure 6: Learned bandwidth evolution in the BLR example.

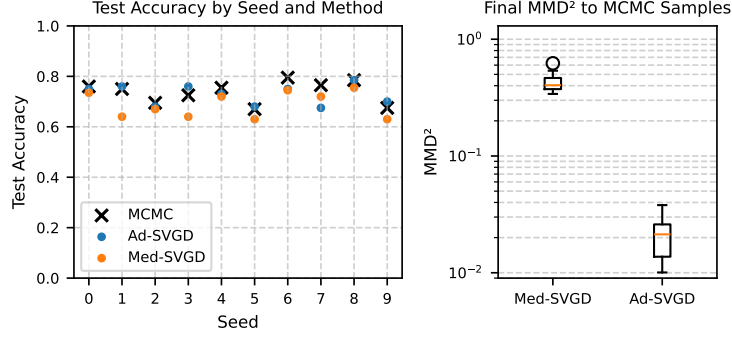


Figure 5: Bayesian logistic regression contribution: SVGD particle quality measured by prediction accuracy (left) and MMD^2 to reference samples (right).

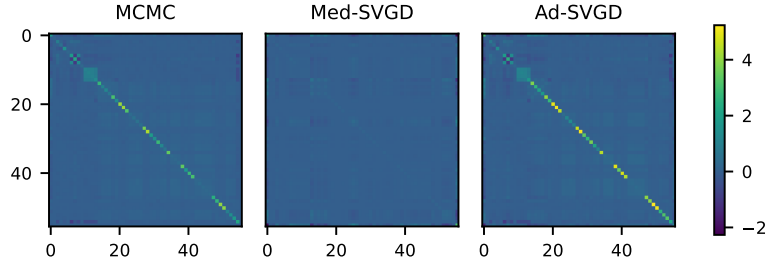


Figure 7: Posterior uncertainty comparison for BLR: covariance matrices of final SVGD particle sets compared with MCMC samples, averaged over 10 seeds.

tainty, whereas Ad-SVGD gives a closer approximation. Figure 6 further shows the learned component-wise bandwidths. Starting from the uninformed initialization of 1 in every component, Ad-SVGD recovers bandwidths that differ from the initialization by several orders of magnitude.

6 Limitations and outlook

The main limitation of the theoretical contribution is its reliance on Assumption 4. Our experiments use the alternating gradient-ascent scheme in Algorithm 1, and ongoing work studies when this scheme satisfies Assumption 4. Thus the current paper provides a convergence analysis conditional on Assumption 4 and empirical evidence that the implemented procedure is effective, but it does not yet prove that the practical gradient-ascent update always satisfies the assumption.

A second limitation is scope: we deliberately analyze the original SVGD dynamics to isolate the kernel-selection contribution. A natural next step is to combine adaptive kernel selection with

recent variants such as sliced SVGD [10], Grassmann SVGD [21], or Stein transport [27].

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A More numerical experiments

A.1 Gaussian mixture models

We revisit the one-dimensional example from Section 5.2 to separate two messages: fixed-bandwidth SVGD is sensitive to h , while adaptive selection is competitive with the median heuristic in this simple setting. For the setup described there, Figure 9 compares final particle histograms with the target density for $h \in \{0.001, 1, 1000\}$. We then compare Med-SVGD and Ad-SVGD across particle counts M , using the final particle distributions after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle distribution and the target for Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As expected, approximation quality improves with the number of particles for both methods. In this low-dimensional case, the two methods are similar and both reach a Wasserstein distance below 0.01 for $N = 500$.

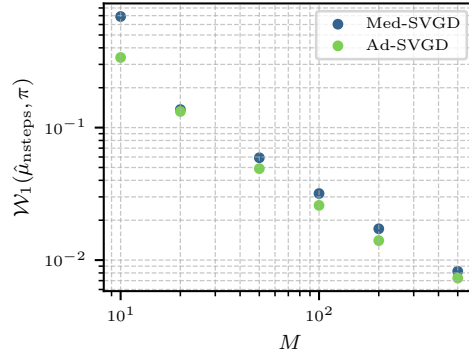


Figure 8: One-dimensional Gaussian mixture comparison: final approximation error as the particle count varies.

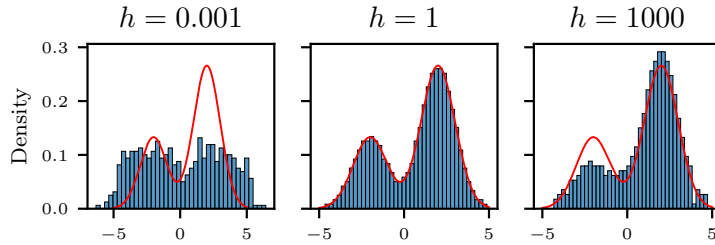


Figure 9: Fixed-bandwidth sensitivity in the one-dimensional mixture: final particle histograms for $h = 0.001, 1, 1000$, with target density π shown in red.

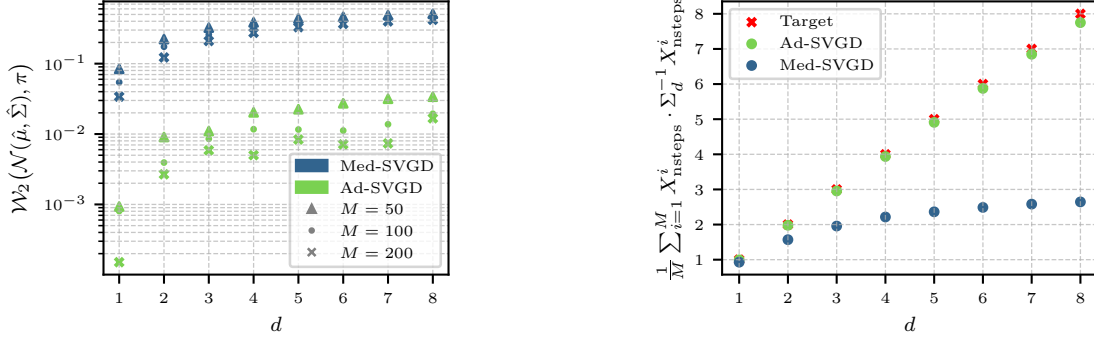
Thus the one-dimensional experiment supports a narrow conclusion: adaptive bandwidth selection works well, but it does not provide a meaningful advantage over the median heuristic when the target is simple and one-dimensional. The value of the contribution becomes more visible in the higher-dimensional tests below.

A.2 Scaling dimension: multivariate normal distribution

To test the contribution under dimensional scaling, we next use Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The initial particle distribution is $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and the algorithms run for 10^4 iterations with step size 0.1. When numerical stability required it, we used a smaller step size and adjusted the number of iterations accordingly.

We compare final-particle approximation quality for Med-SVGD and Ad-SVGD as d increases. Because the Wasserstein 2-distance between Gaussian distributions has an explicit formula (see

[29]), we measure sample quality by the Wasserstein 2-distance between π_d and $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical particle mean and covariance [29, Section 3.2]. Following the terminology of the Python Optimal Transport library [9], used for the calculations, we call this the *Bures Wasserstein distance*. Figure 10a shows this measure for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as dimension increases, and the adaptive method significantly outperforms Med-SVGD for all d .



(a) Bures Wasserstein approximation error for $M \in \{50, 100, 200\}$. (b) Uncertainty-sensitive χ^2 -test statistic for $M = 200$.

Figure 10: Dimensional-scaling contribution: Med-SVGD and Ad-SVGD on multivariate normal targets of increasing dimension.

The Bures Wasserstein distance gives limited weight to the smallest-variance components of π_d , so we also use an uncertainty-sensitive statistic. For $X \sim \mathcal{N}(0, \Sigma_d)$, the statistic $X \cdot \Sigma_d^{-1} X$ is χ^2 -distributed with d degrees of freedom and expected value d (cf. [34, p. 27]). We compute its mean on the particle set with $M = 200$ and compare it with d in Figure 10b. As dimension grows, the Med-SVGD statistic moves farther from the true expectation, while the Ad-SVGD statistic remains comparatively close. This result directly supports the contribution claim that adaptive kernel selection mitigates higher-dimensional variance underestimation.

Tables 1 and 2 provide component-level evidence for $M = 200$ and varying d . Each table lists particle marginal variances against the target marginal variances in the first row: Table 1 reports Med-SVGD and Table 2 reports Ad-SVGD. Ad-SVGD closely matches the target marginal variances across components, whereas Med-SVGD substantially underestimates uncertainty, consistent with Figure 10b.

Finally, for $M = 200$ and $d = 8$, we visualize marginal particle distributions after normalizing each component by the inverse target marginal standard deviation, equivalently multiplying the i -th component by i . Under this normalization, each target marginal becomes standard normal. Figures 11 and 12 show the resulting histograms for Med-SVGD and Ad-SVGD, with a standard Gaussian density for comparison. Ad-SVGD captures all marginals well, while Med-SVGD again underestimates uncertainty. The same pattern appears in Figures 13 and 14, which compare normalized marginal particle quantiles with standard-normal target quantiles.

Table 1: Variance-collapse diagnostic for Med-SVGD: marginal variances of final particles for d -dimensional examples with $M = 200$, with target variances of π_d shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Variance-recovery diagnostic for Ad-SVGD: marginal variances of final particles for d -dimensional examples with $M = 200$, with target variances of π_d shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

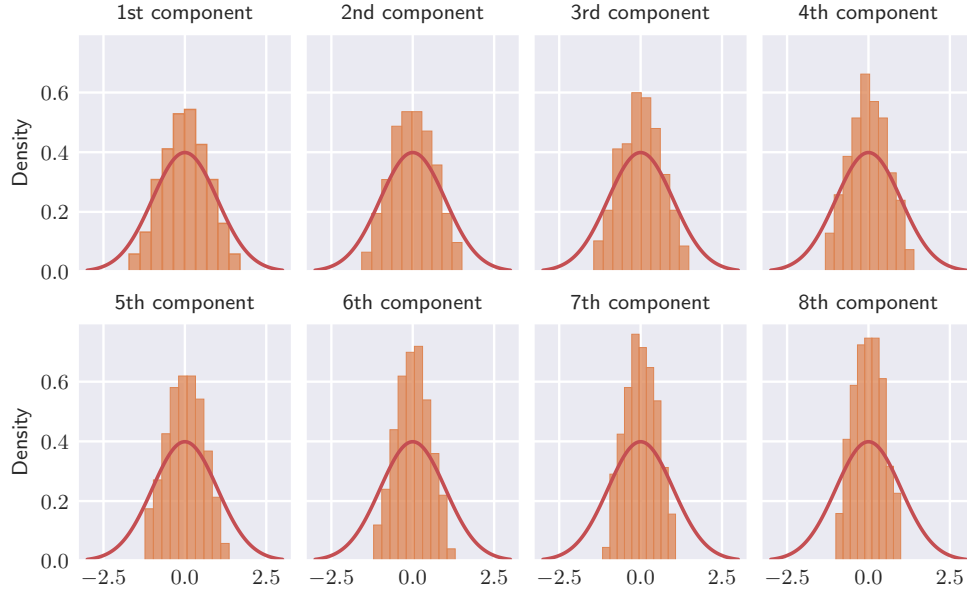


Figure 11: Med-SVGD marginal histograms for the eight-dimensional example with $M = 200$, rescaled by target marginal variances; standard Gaussian density shown for comparison.

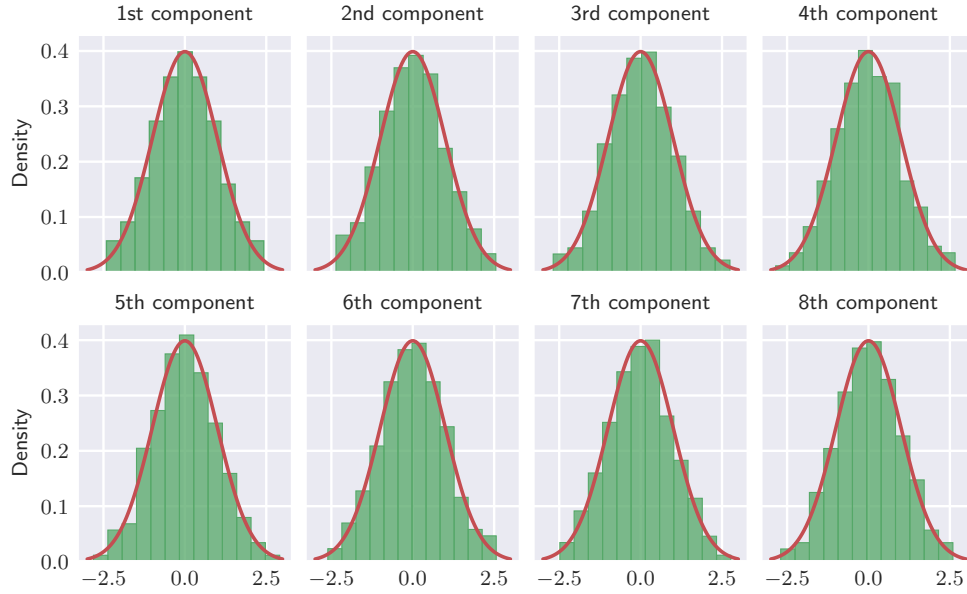


Figure 12: Ad-SVGD marginal histograms for the eight-dimensional example with $M = 200$, rescaled by target marginal variances; standard Gaussian density shown for comparison.

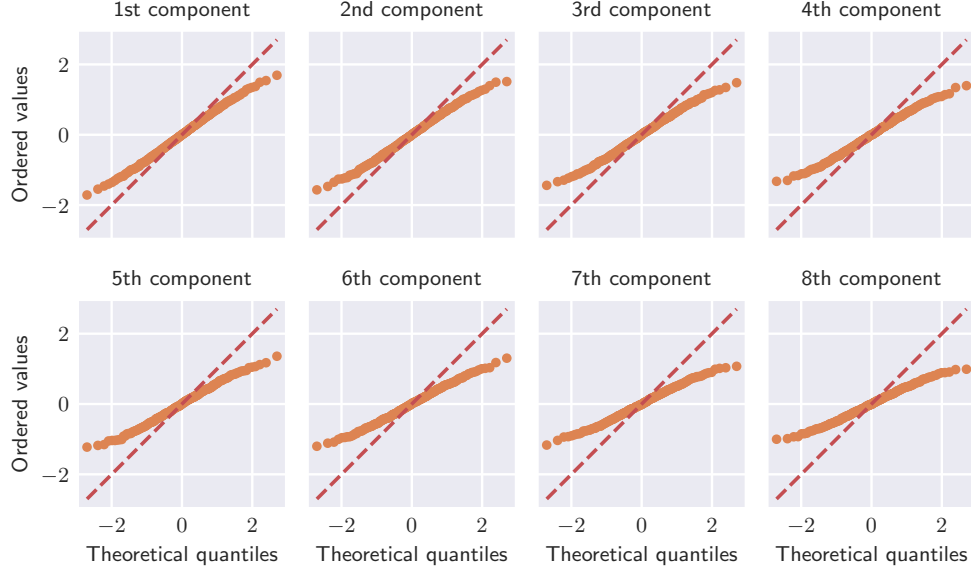


Figure 13: Med-SVGd Q-Q plots for normalized marginals in the eight-dimensional example with $M = 200$; the standard-normal reference line has slope 1 and passes through the origin.

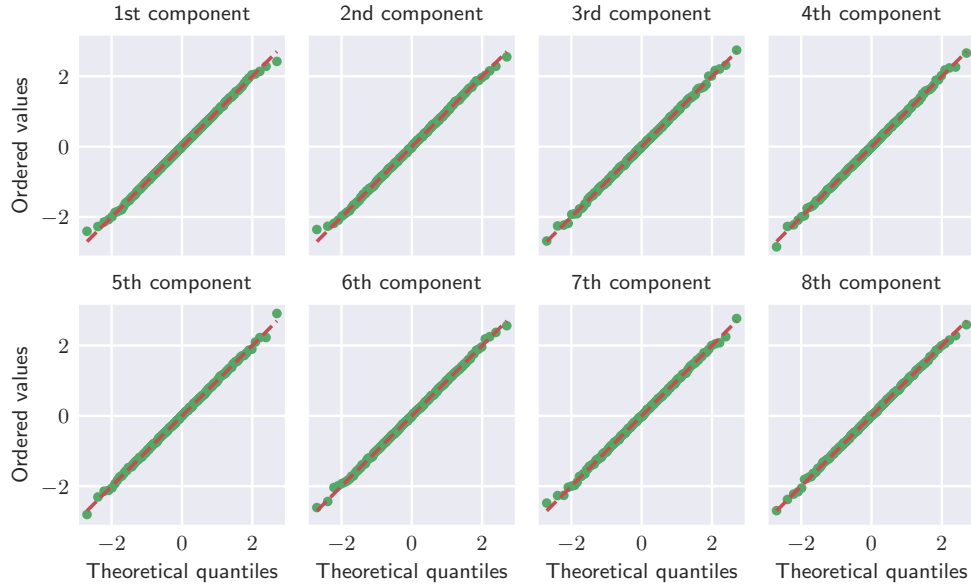


Figure 14: Ad-SVGd Q-Q plots for normalized marginals in the eight-dimensional example with $M = 200$; the standard-normal reference line has slope 1 and passes through the origin.

A.3 Gaussian process inference

The Gaussian process (GP) experiment provides a second dimensional-scaling check for the adaptive-kernel contribution, using the inference task proposed in [31]. We consider a GP on $[0, 1]$ represented by the truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ and the coefficients are independent Gaussian variables $x_k \sim \mathcal{N}(0, k^{-2})$. The process is observed at N_y equispaced points in $[0, 1]$, and the task is to infer the coefficients x_k . For fixed N_x and N_y , this is an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ and $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal entries k^{-2} , $k = 1, \dots, N_x$, and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

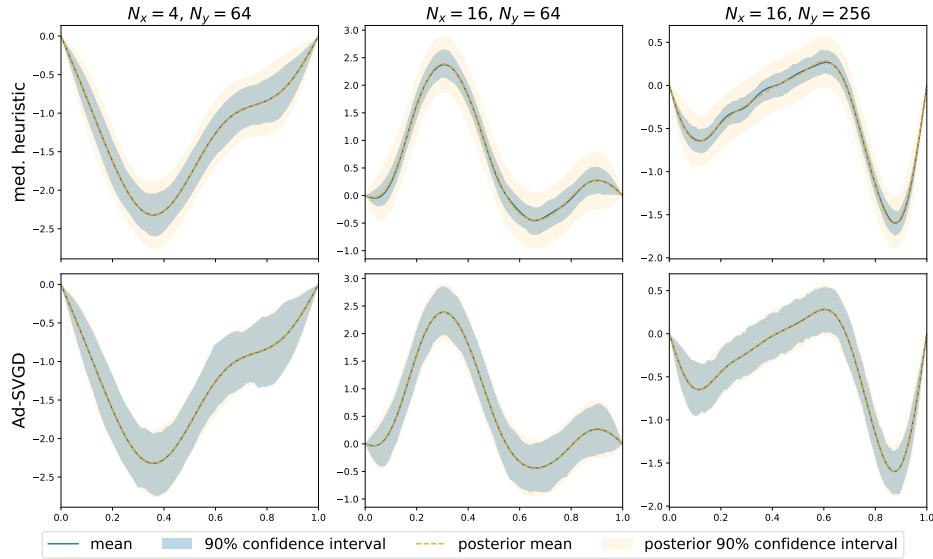


Figure 15: GP inference comparison: estimated processes from median-heuristic SVGD and Ad-SVGD with $M = 100$ particles for different model configurations, compared with the posterior.

We sample from the posterior $X | Y = \bar{y}$ with SVGD and compare the median heuristic with the adaptive bandwidth rule. Median-heuristic SVGD uses kernels $k(x, y) = \exp(-\|x - y\|_1/h)$, while Ad-SVGD uses product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with $h = (h_1, \dots, h_{N_x})$, giving each dimension its own bandwidth. Following [18], the $N_x = 16$ scenarios use an Adagrad variant for step-size control.

Figure 15 compares the inferred processes with the posterior for $N_x = 4, N_y = 64$, $N_x = 16, N_y = 64$, and $N_x = 16, N_y = 256$. As dimension increases, median-heuristic SVGD underestimates posterior variance, whereas Ad-SVGD gives a closer uncertainty approximation. The pattern is consistent across the two observation counts. Table 3 quantifies the same contribution by comparing the trace of the particle covariance with the true posterior trace: the median heuristic severely underestimates uncertainty, while Ad-SVGD captures the variance more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Posterior-variance recovery in GP inference: trace of final-particle covariance ($M = 100$ particles) compared with the theoretical posterior for different configurations, averaged over 25 runs.

B Refined theoretical analysis

This appendix states the refined theoretical contribution behind the summary in the main text. Under a Stein log-Sobolev inequality, which relates KL divergence to KSD, the adaptive KSD convergence result can be strengthened to KL bounds. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it may hold. In the adaptive kernel-selection setting, the condition can be stated over the kernel class and only requires that the maximal KSD controls the KL divergence, which is satisfied if one suitable kernel in the class provides that control. We use the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Under this property, the analysis quantifies convergence directly in KL divergence. A useful feature of the result is that it also bounds $\mathcal{F}(\mu_n)$ when the KSD maximization is solved only to a fixed accuracy $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. As in the proof of Theorem 3, Assumption 5 and the stated conditions on α, γ give

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$ gives the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

The three stated approximation regimes then follow separately:

1. If $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows from the geometric series bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. If $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

using the geometric-series formula. Similarly, when $q < \rho$, we directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, when $q = \rho$, bounding $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ gives the claim by the same argument.

3. If $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho} .
\end{aligned}$$

This uses $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□